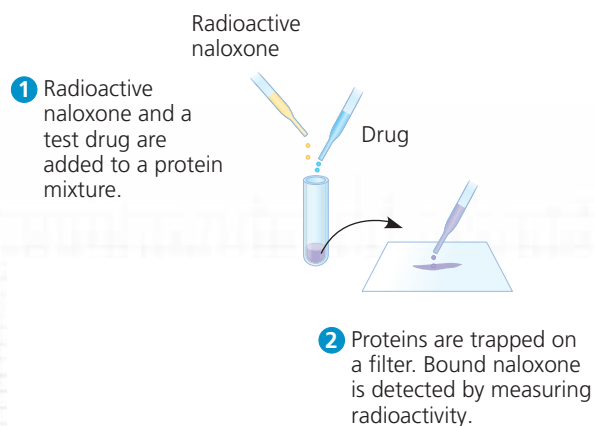


Interpreting Data Values Expressed in Scientific Notation

Does the Brain Have Specific Protein Receptors for Opiates? Researchers were looking for opiate receptors in the mammalian brain. Knowing that the drug naloxone blocks the analgesic effect of opiates, they hypothesized that naloxone acts by binding tightly to brain opiate receptors without activating them. In this exercise, you will interpret the results of an experiment to test this hypothesis.

How the Experiment Was Done The researchers incubated radioactive naloxone with a protein mixture prepared from rodent brains. If the mixture contained opiate receptors or other proteins that could bind naloxone, the radioactivity would stably associate with the mixture. To determine whether the binding was due to specific opiate receptors, they tested other drugs, opiate and non-opiate, for their ability to block naloxone binding.



to relieving pain, they reduce urine output, decrease respiration, and produce euphoria as well as other emotional effects. Because opiates (drugs such as morphine and heroin) bind to the same receptor proteins as endorphins, opiates mimic endorphins and produce many of the same physiological effects (see Figure 2.16). In the **Scientific Skills Exercise**, you can interpret data from an experiment designed to search for opiate receptors in the brain.

Gases

Some vertebrate neurons release dissolved gases as neurotransmitters. In human males, for example, certain neurons release nitric oxide (NO) into the erectile tissue of the penis during sexual arousal. The resulting relaxation of smooth

Data from the Experiment

Drug	Opiate	Lowest Concentration That Blocked Naloxone Binding
Morphine	Yes	$6 \times 10^{-9} M$
Methadone	Yes	$2 \times 10^{-8} M$
Levorphanol	Yes	$2 \times 10^{-9} M$
Phenobarbital	No	No effect at $10^{-4} M$
Atropine	No	No effect at $10^{-4} M$
Serotonin	No	No effect at $10^{-4} M$

Data from C. B. Pert and S. H. Snyder, Opiate receptor: demonstration in nervous tissue, *Science* 179:1011–1014 (1973).

INTERPRET THE DATA

- The data above are expressed in scientific notation: a numerical factor times a power of 10. Remember that a negative power of 10 means a number less than 1. For example, $10^{-1} M$ (molar) can also be written as $0.1 M$. Write the concentrations in the table for morphine and atropine in this alternative format.
- Compare the concentrations listed in the table for methadone and phenobarbital. Which concentration is higher? By how much?
- Would phenobarbital, atropine, or serotonin have blocked naloxone binding at a concentration of $10^{-5} M$? Explain.
- (a) Which drugs blocked naloxone binding in this experiment?
(b) What do these results indicate about the brain receptors for naloxone?
- When the researchers instead used tissue from intestinal muscles rather than brains, they found no naloxone binding. What does that suggest about opiate receptors in mammalian muscle?

➔ **Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

muscle in the blood vessel walls of the spongy erectile tissue allows the tissue to fill with blood, producing an erection. The erectile dysfunction drug Viagra works by inhibiting an enzyme that terminates the action of NO.

Unlike most neurotransmitters, NO is not stored in cytoplasmic vesicles but is instead synthesized on demand. NO diffuses into neighboring target cells, produces a change, and is broken down—all within a few seconds. In many of its targets, including smooth muscle cells, NO works like many hormones, stimulating an enzyme to synthesize a second messenger that directly affects cellular metabolism.

Although the gas carbon monoxide (CO) is a deadly poison if inhaled, vertebrates produce small amounts of CO as a neurotransmitter. For example, CO synthesized